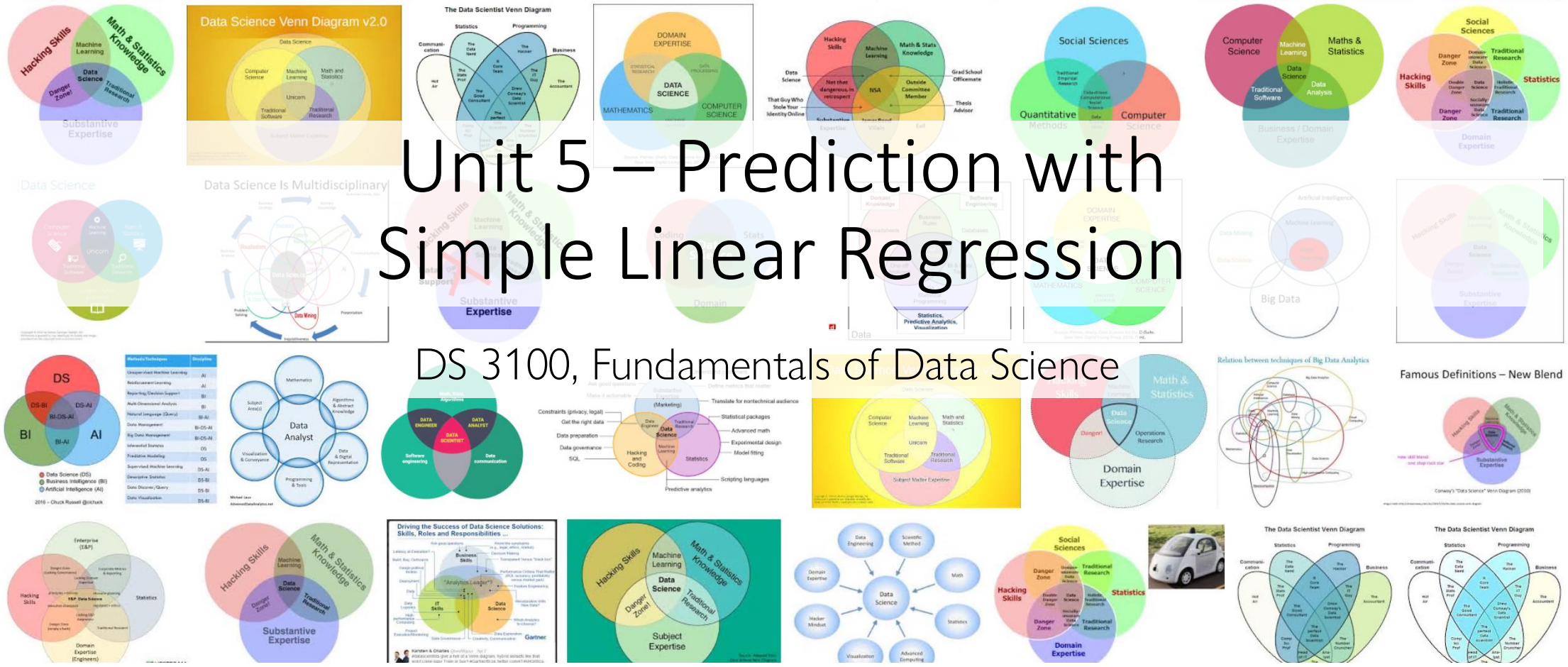


Unit 5 – Prediction with Simple Linear Regression

DS 3100, Fundamentals of Data Science

VANDERBILT UNIVERSITY

Objectives

- Moving beyond correlation for Linear Associations
- Fitting a Line to Data
- Objective functions for Linear Regression
- Ordinary Least Squares

Regression

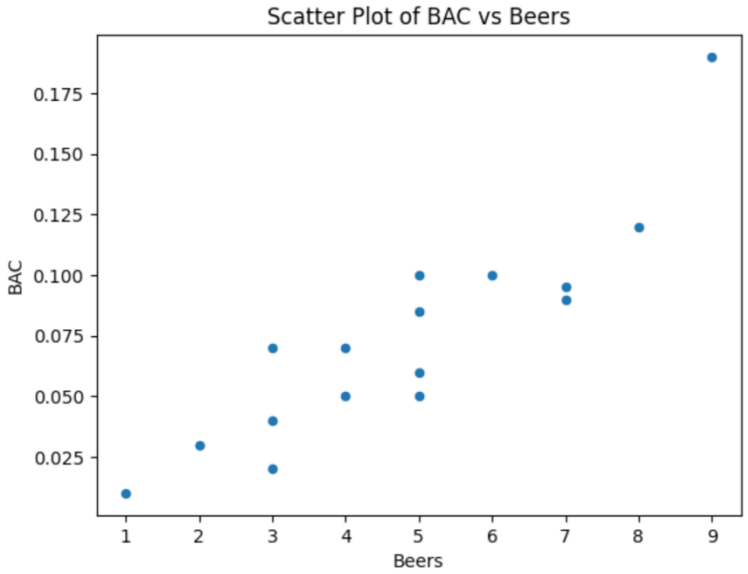
- Regression is a statistical framework for describing and modeling how an outcome varies with one or more predictors.
- Predictive problem: Rather than asking only whether two variables are associated, regression lets us ask
  - How does the expected value of an outcome change as a predictor changes? For example:
    - How does house price change with the size of a house?
    - How does exam performance change with hours spent studying?
    - How does life expectancy vary with health expenditure?
    - How does energy consumption change with outside temperature?
- We begin with the simplest case: one quantitative outcome and one predictor.
  - **Simple linear regression** asks whether the *expected value of Y* can be reasonably represented as a linear function of X: $y = mx + c$

Setting the Regression scene

A simple example to set the premise for linear equations and Simple Linear Regression.

In a perfect world, we would be able to predict someone's BAC just by knowing the number of beers they drank using a linear equation. However, there are very few times when such linear relationships are perfect. Rather than finding an exact solution, we typically want to find something that is ***close enough*** (an *approximate* solution). Plotting the data shows a general linear trend which is reassuring.

```
bac_data.plot.scatter( x='Beers', y='BAC',title='Scatter Plot of BAC vs Beers')
plt.show()
```





A Linear Equation

This is a linear equation in two dimensions, representing a line fit.

$$y = mx + c \rightarrow BAC = \beta_0 + \beta_1 \times Beers$$

A few key terms:

- Explanatory Variable: # of Beers; known from our dataset.
  Also called predictor or feature.
- Response Variable: BAC; what we want to predict
- β_0 is the intercept of the line that represents the linear relationship between predictor and response.
  - expected value of Y when al values of X are zero
- β_1 is the slope of the line.
  - number of units Y changes with respect to 1 unit change in X

$$0.100 = \beta_0 + \beta_1 \times 5$$
$$0.030 = \beta_0 + \beta_1 \times 2$$
$$\vdots$$
$$0.090 = \beta_0 + \beta_1 \times 7$$
$$0.010 = \beta_0 + \beta_1 \times 1$$
$$0.050 = \beta_0 + \beta_1 \times 4$$

Given our dataset, we are trying to fit *a system of equation*.

Solving a linear system of equations

Modern computational packages represent them in vectorial form. $X\vec{\beta}$ is a *linear combination of the columns* of the matrix X. We want to find an estimate of $\vec{\beta}$ that is the *best approximation* of $\vec{y}$

$$
\begin{aligned}
0.100 &= \beta_0 + \beta_1 \times 5 \\
0.030 &= \beta_0 + \beta_1 \times 2 \\
&\vdots \\
0.090 &= \beta_0 + \beta_1 \times 7 \\
0.010 &= \beta_0 + \beta_1 \times 1 \\
0.050 &= \beta_0 + \beta_1 \times 4
\end{aligned}
$$

We can re-express the system of linear equations as follows:

$$X\vec{\beta} = \vec{y}$$

$$
\begin{bmatrix} 1 & 5 \\ 1 & 2 \\ 1 & 9 \\ & \vdots \\ 1 & 1 \\ 1 & 4 \end{bmatrix} \vec{\beta} = \begin{bmatrix} 0.100 \\ 0.030 \\ 0.190 \\ \vdots \\ 0.010 \\ 0.050 \end{bmatrix} \text{; where } \vec{\beta} = \begin{bmatrix} \beta_0 \\ \beta_1 \end{bmatrix}
$$

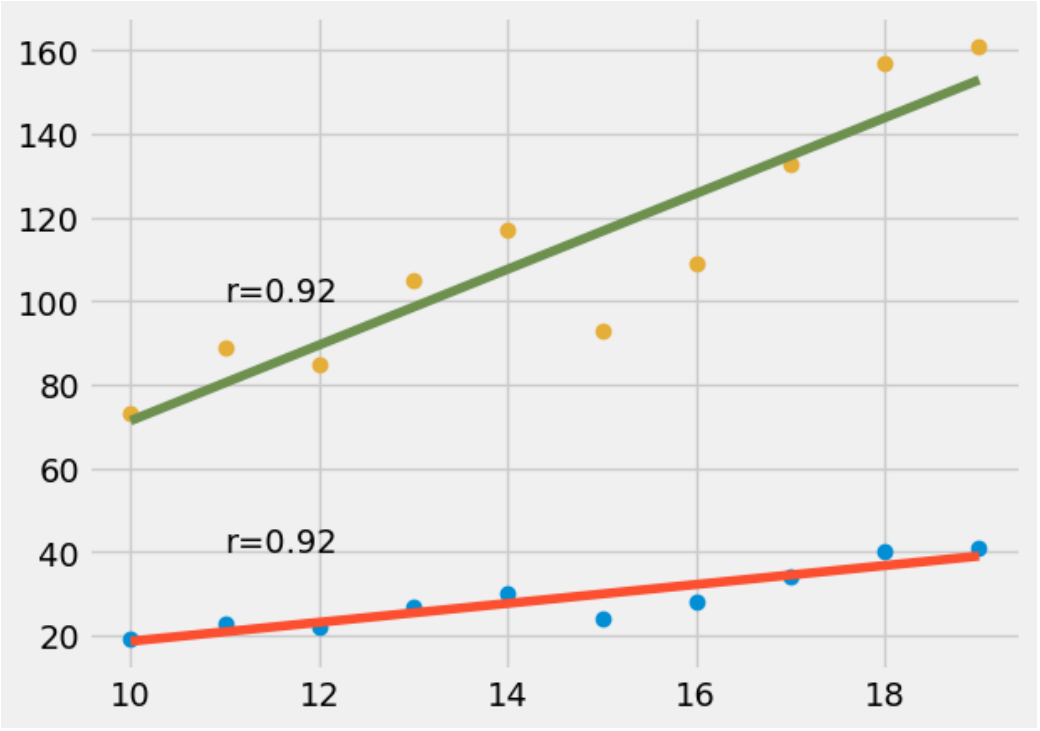
Note that we inject an extra column of all **1**s alongside the actual feature column to make our matrix operation consistent with the fact that the parameter vector $\vec{\beta}$ includes a non-zero intercept in most cases of Simple Linear Regression.

Why correlation isn't the whole story

- Correlation tells us about the strength of a linear relationship
  - Correlation coefficient r can still be insufficient in establishing the relationship between two variables

Same correlation coefficient between these two data spreads.
But exact relationship is different.

How do you find the linear relationship between correlated variables?
We fit a line to the data!



Fitting a line to the data

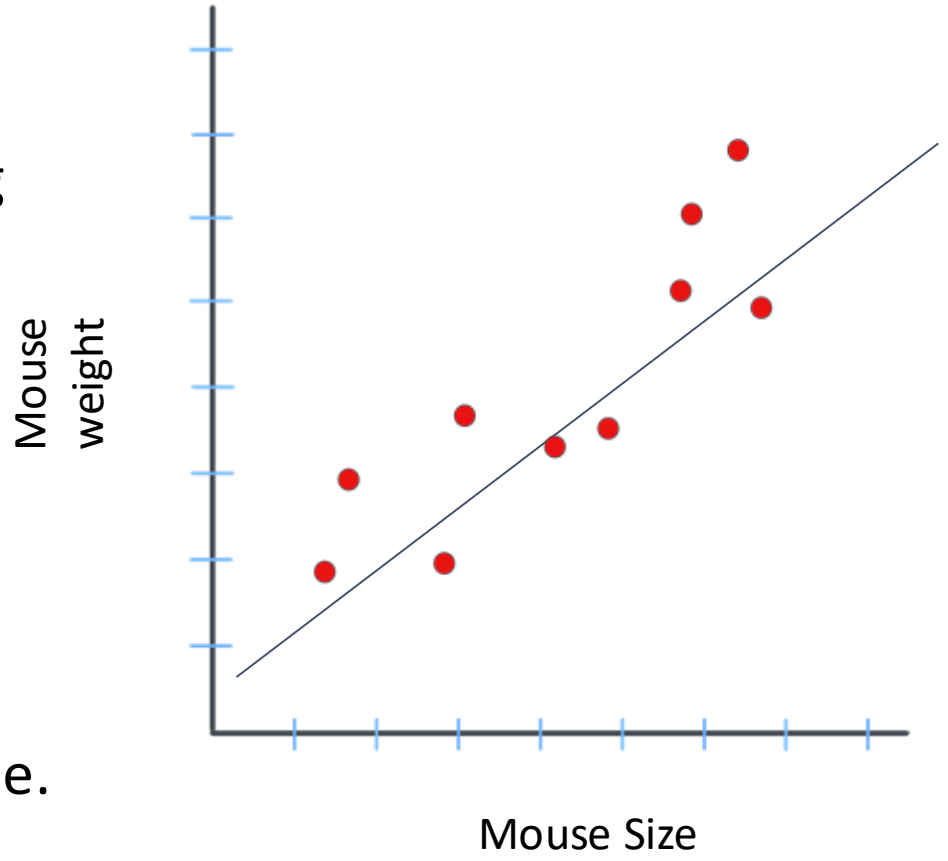
Suppose we have collected some data on mice size, and we are trying to predict mice weight from it.

That is a regression problem which would be trying to make quantitative predictions.

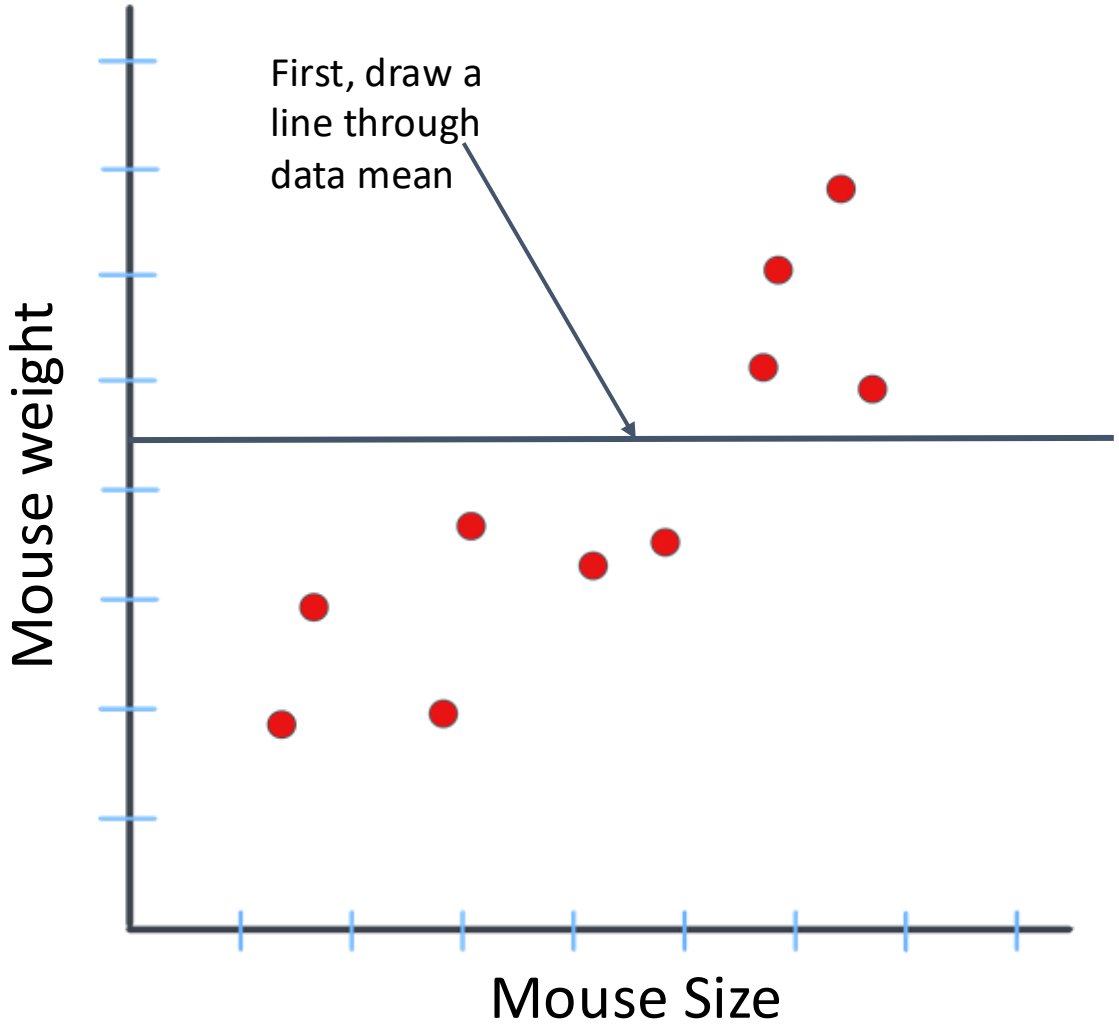
A simple way to imagine this is to fit a line on the experimental data s.t. the equation of the line produces mouse weight when we enter the mouse size as input.

We need an algorithm for finding the best fitted line.

The linear relationship can be written as $y = \beta_0 + \beta_1 x$



Fitting a line to data : Baseline

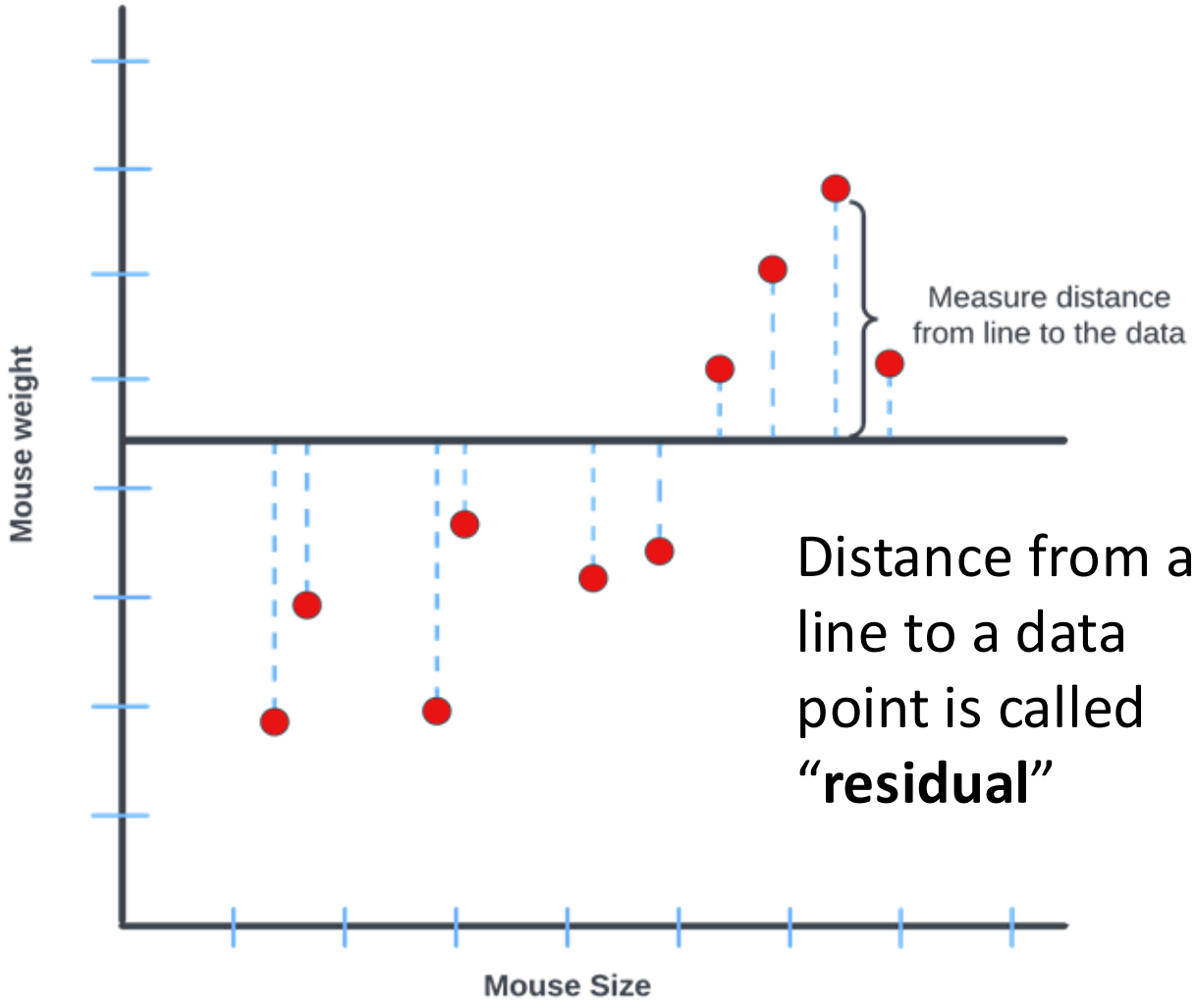


Fitting a line to data : Residuals

Next an important part of line fitting is to measure the distance between a datapoint and the predicted value.

This is essentially the length of the **orthogonal distance**, dropped from the point to the fitted line.

This is called a **residual** for a prediction, and it will be an important part of optimizing the fitted line.



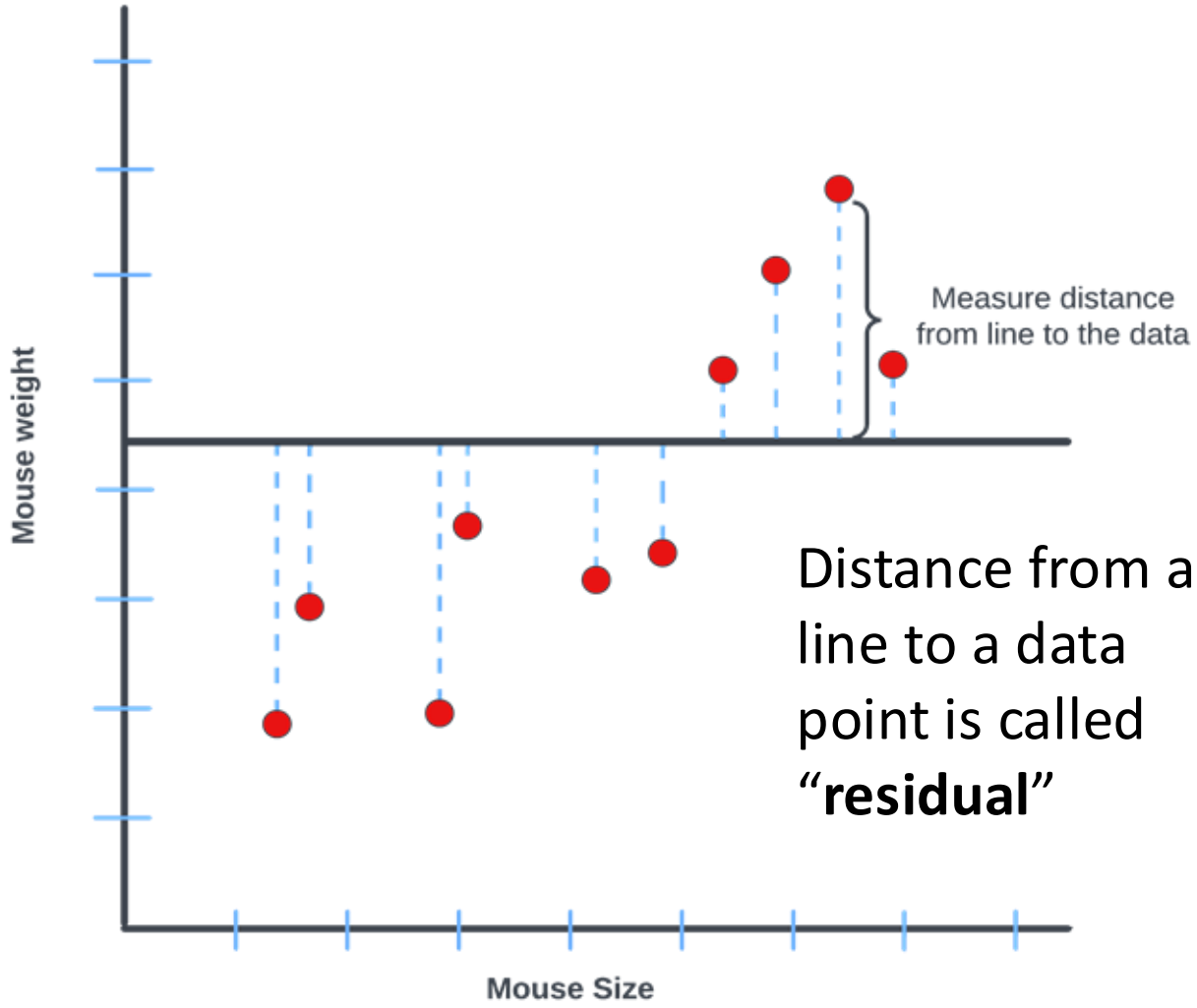
Distance from a line to a data point is called "**residual**"

Fitting a line to data : Residuals

Residuals could be positive or negative depending on if the y-value is above or below the line.

Directly summing positive and negative residuals would typically lead to a misleading aggregate error.
Makes the fit appear better than it is!

Sum of absolute residuals or squared residuals fixes this.
Later is preferred to facilitate a convex function for optimization



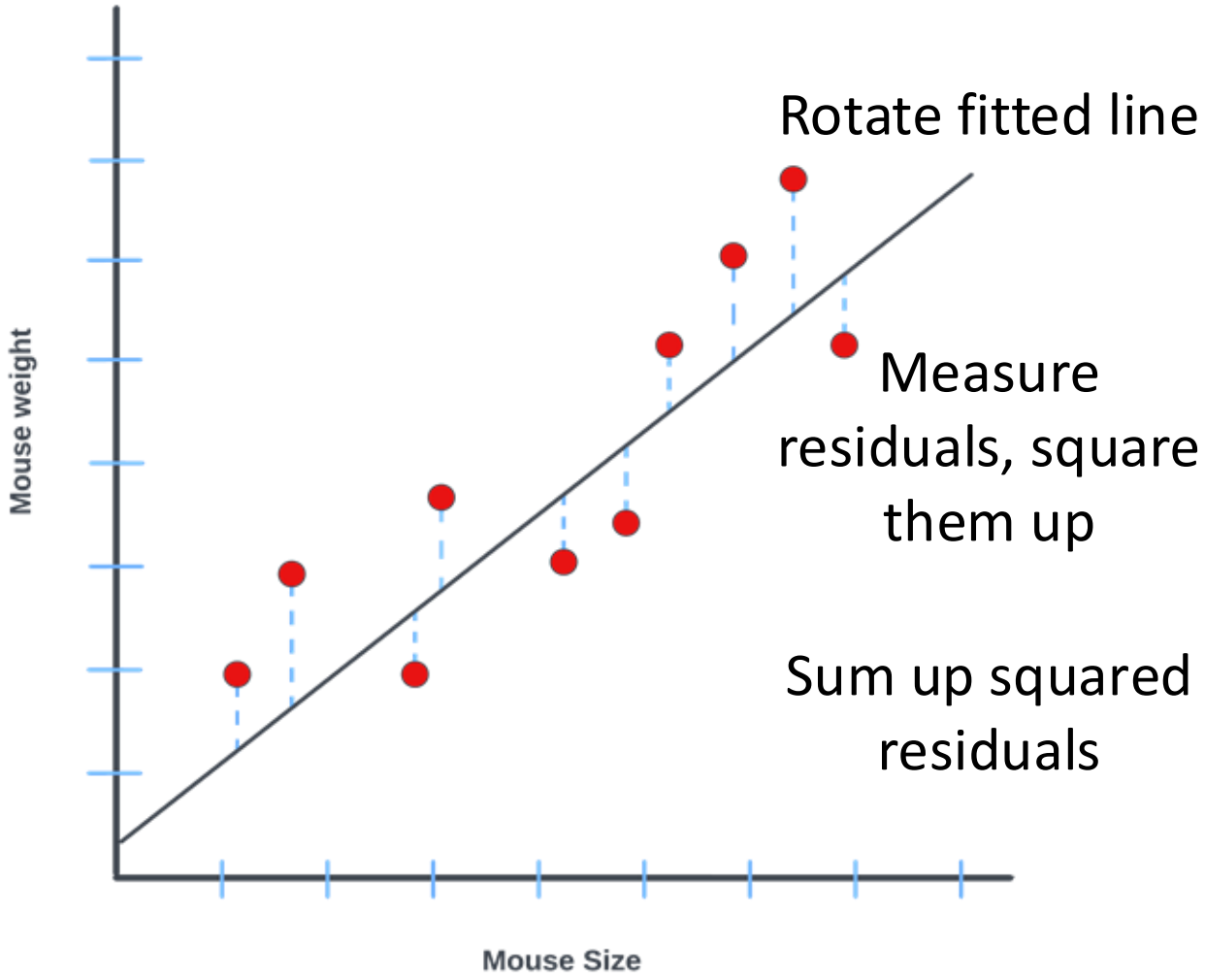
Distance from a line to a data point is called "**residual**"

Fitting a line to data : Ordinary Least Squares

Think of the fitting as an *iterative process* where each iteration essentially consists of 3 steps:

1. Start with a line
2. Measure residuals for every predicted point and square them up
3. Sum up the squared residuals to obtain the **squared loss** corresponding to the current fit

Next, we rotate the line slightly and repeat all the three steps that I just described.

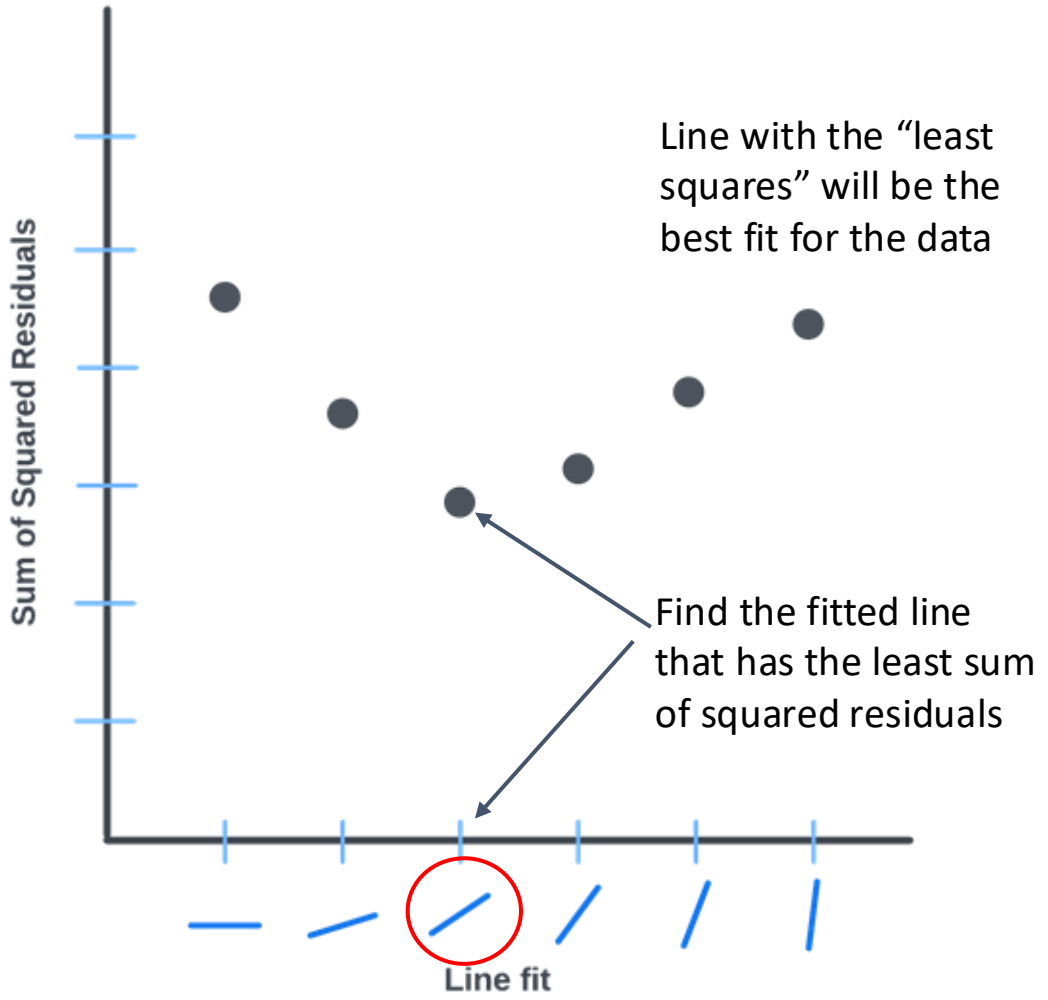


Fitting a line to data : Ordinary Least Squares

Then we plot the sum of squared residuals for every line we attempt to fit on the data.

The optimal fitted line for the data would be the one that yields the lowest sum of squared residuals.

Hence this method is called the **least squares method**.



Ordinary Least Squares Regression

- Decide the best parameter values for the regression line, $\hat{y} = \beta_0 + \beta_1 x$
  - β_0 is the intercept parameter and β_1 is the slope parameter
- Optimization: Pick the line that minimizes the sum of squared residuals (RSS)
  - Residual: Difference between a prediction and an observed response. $e_i = y_i - \hat{y}_i$
  - Objective function: This is also more commonly known as a loss function.
  - Minimize RSS to find the *line of best fit*.
- Parameters can be mathematically derived in a *closed form solution* of the optimization problem as: $\widehat{\beta_1} = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^{n}(x_i - \bar{x})^2} = r \times \frac{s_y}{s_x}$ and $\widehat{\beta_0} = \bar{y} - \widehat{\beta_1}\bar{x}$
  - s_x and s_y are standard deviations of variables x and y from our samples
  - $\bar{x}$ and $\bar{y}$ are the respective means of x and y
- Together these parameters give us the equation of the least squares line for simple linear regression

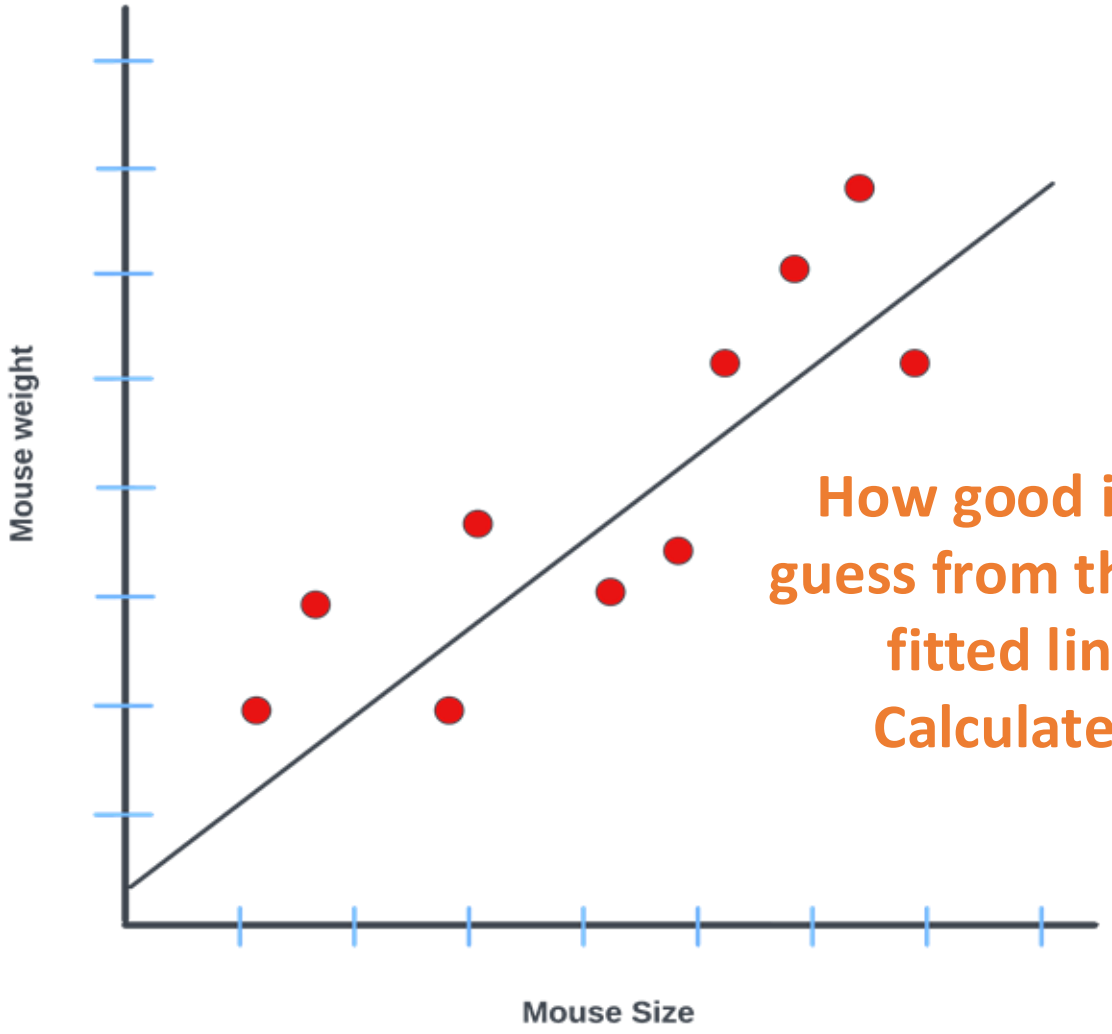
Goodness of fit

So now we have:

- Defined a *loss function* to optimize for linear regression.
- Specified how to estimate the parameters that defines the optimal line (hyperplane).

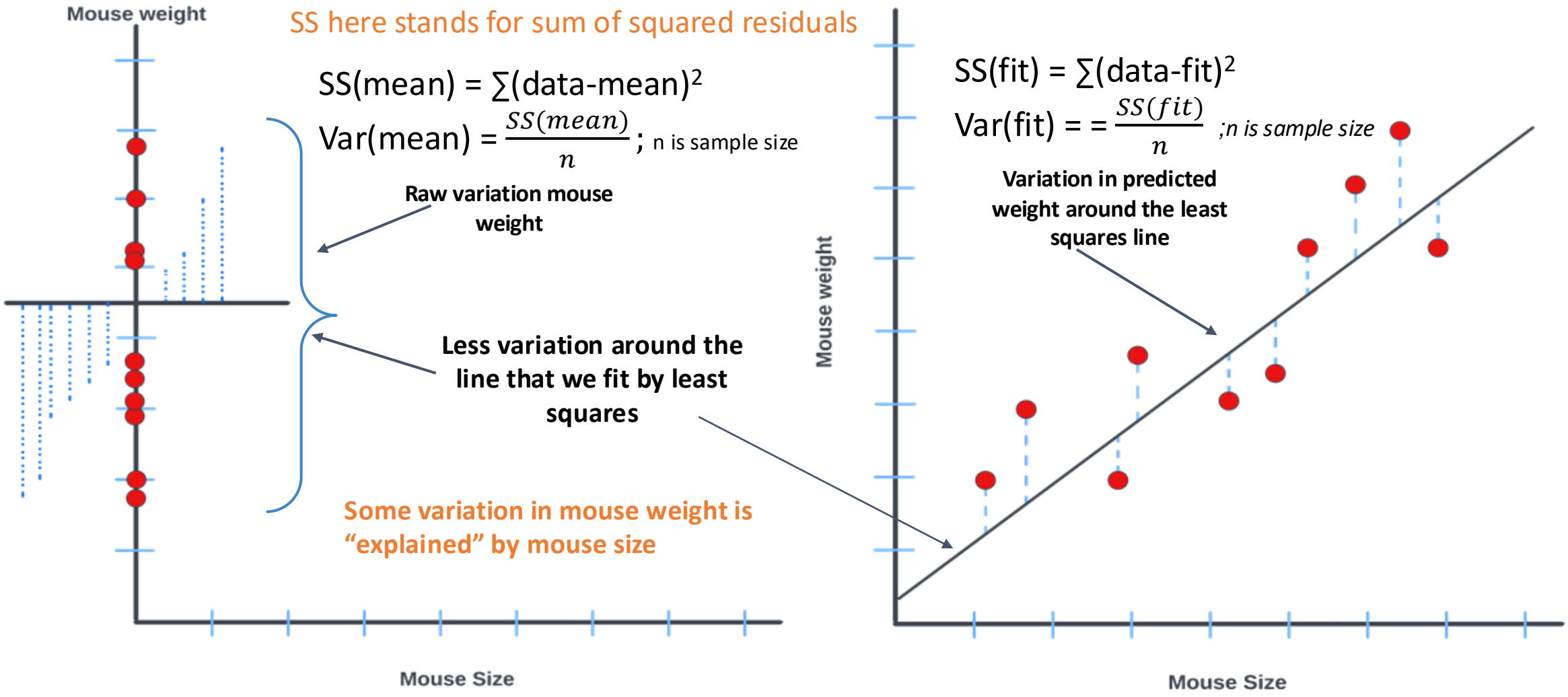
We now have a measure of error. Next question is how to gauge the **goodness of fit** we achieve with it? We do this by calculating R^2 otherwise known as the *coefficient of determination*.

$$R^2 = (r)^2$$

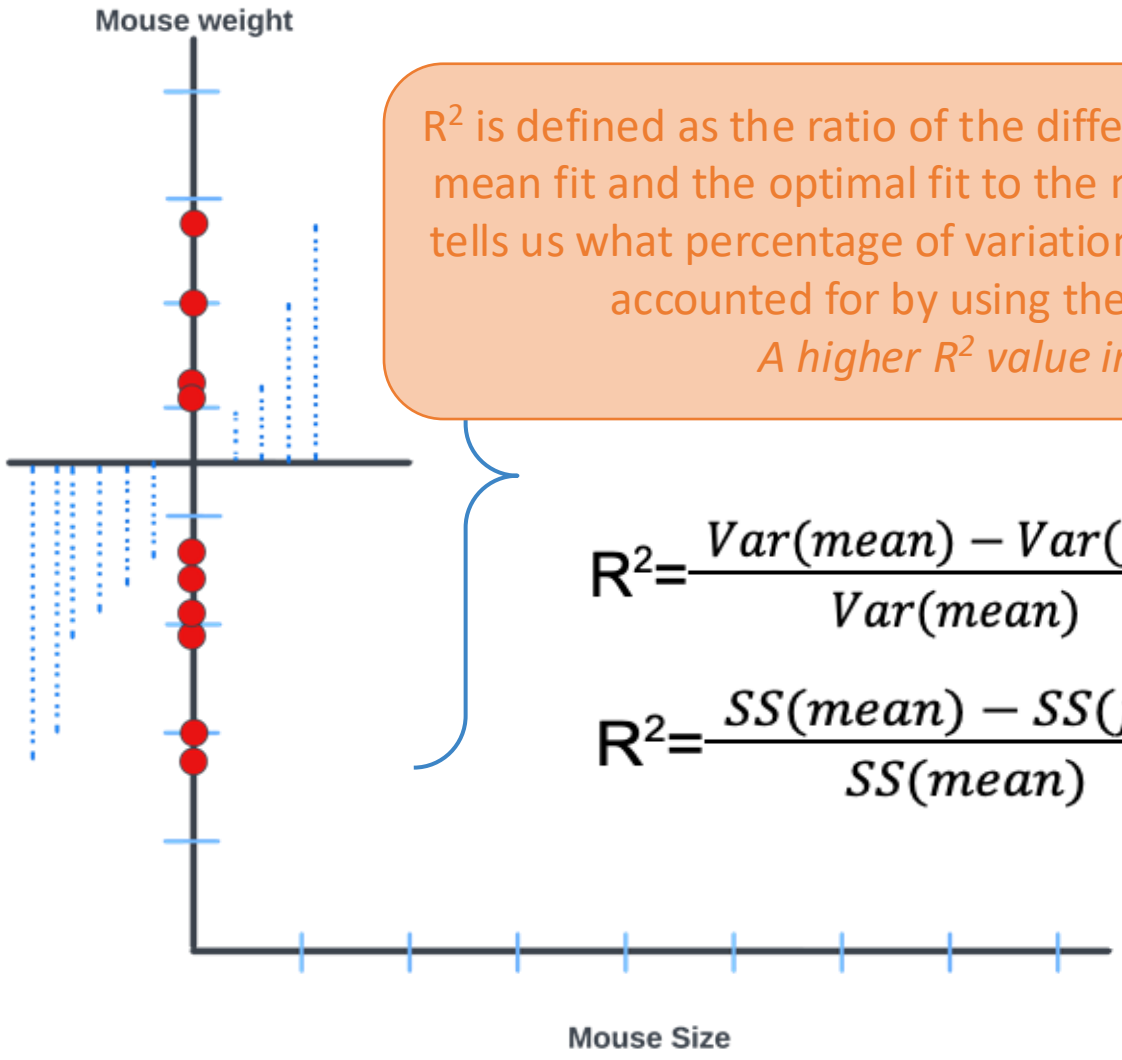


How good is the guess from the best fitted line? Calculate R^2

Goodness of fit: R-squared



Goodness of fit

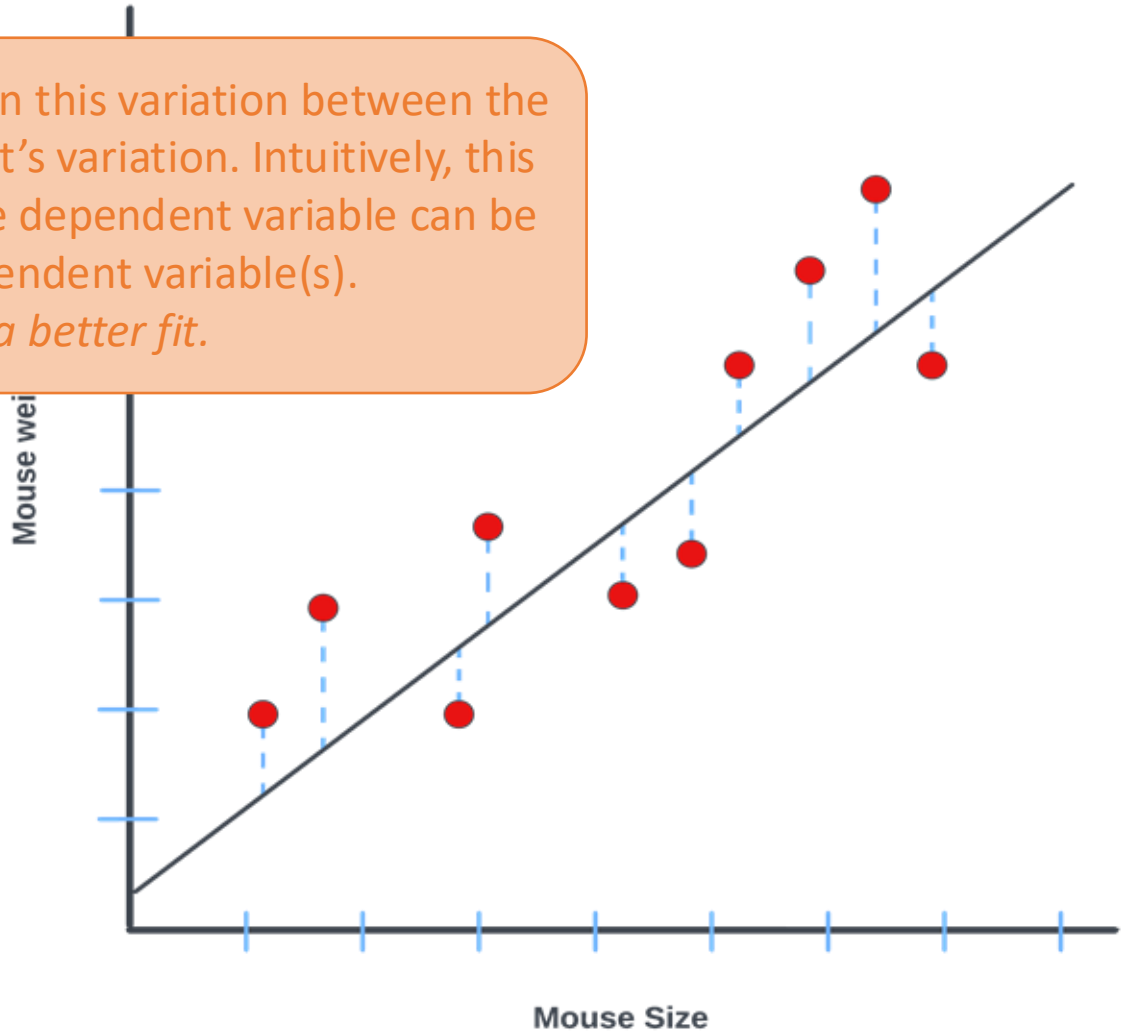


R^2 is defined as the ratio of the difference in this variation between the mean fit and the optimal fit to the mean fit's variation. Intuitively, this tells us what percentage of variation of the dependent variable can be accounted for by using the independent variable(s).

A higher R^2 value implies a better fit.

$$R^2 = \frac{Var(mean) - Var(fit)}{Var(mean)}$$

$$R^2 = \frac{SS(mean) - SS(fit)}{SS(mean)}$$



Overall significance of fit

- What if you only had two values?
  - You can always fit a line through any two points → You can always get a 100% R^2
  - Doesn't mean the model learnt anything.
- Need a way to determine of the R^2 value is statistically significant → A hypothesis test using p-values!
  - Overall Test or **F-test**: A test of slopes.
    It compares hypothesized model to intercept-only model, whether predictions are better than always predicting the mean. For simple linear regression,
    - Null hypothesis $H_0: \beta = 0$
    - Alternative hypothesis $H_A: \beta \neq 0$
  - For this test, we calculate the **F-statistic** and then refer to the F-distribution.
    - F distribution table

Statistical Significance of Fit

General equation to check if R^2 is significant

$$F = \frac{[SS(mean) - SS(fit)]/(p_{fit} - p_{mean})}{SS(fit)/(n - p_{fit})}$$

$$F = \frac{variation\ in\ mouse\ weight\ explained\ by\ size}{variation\ in\ mouse\ weight\ not\ explained\ by\ size}$$

- p_{mean} : number of parameters in the fit line; *1 for SLR*
- p_{fit} : number of parameters in the mean line; *2 for SLR*

Why divide SS(fit) by n-p_{fit} instead of just n?

More parameters you have in your equation the more data points you need to estimate your fit → less residual degrees of freedom

Is this F statistically significant?

A good fit generates a high F value!

- Get the estimate of F for a fit on original data: F_{obs}
- Refer to the sampling distribution, F distribution.
  - Strictly positive and right skewed
  - The F-test is always a one-tailed (right-tailed) test.
- Two degrees of freedom
  - $df_1 = p_{fit} - p_{mean} = \#\ of\ features(k)$
  - $df_2 = n - p_{fit} = n - k - 1$
- The estimated F statistic must be *greater than* the critical value F_{df_1, df_2} for statistical significance at expected significance level α.

Your turn

Two regression models are fitted to two different datasets. In both cases, the fitted model has the same residual sum of squares: $SS(fit) = 100$.
Which fitted model provides a better fit relative to simply predicting the mean?

	SS(mean)	SS(fit)
Dataset A	125	100
Dataset B	500	100

A simple linear regression is fitted to $n = 20$ observations.
You obtain: $SS(mean) = 500, SS(fit) = 300$. Calculate the F-statistic and determine whether the regression provides a statistically significant improvement over the mean-only model at $\alpha = 0.05$.